Analysis Tutorial 3

Problem 7

(a) ① Clearly $\phi \subseteq \mathbb{N}$, and $|\phi| = 0 < \infty$ so $\phi \in \mathcal{A}$

② Let $A, B \in \mathcal{A}$

If $|A| < \infty$, then $A \setminus B \subseteq A \Rightarrow |A \setminus B| < \infty$

If $|\bar{B}| < \infty$ then $A \setminus B = A \cap \bar{B} \subseteq \bar{B} \Rightarrow |A \setminus B| < \infty$

If $|\bar{A}| < \infty$ and $|B| < \infty$, then $\overline{A \cap \bar{B}} = \bar{A} \cup B$

and $|\bar{A} \cup B| < \infty$

So $A \setminus B \in \mathcal{A}$

③ Let $A, B \in \mathcal{A}$. If either $|\bar{A}| < \infty$ or $|\bar{B}| < \infty$, then $|\overline{A \cup B}| = |\bar{A} \cap \bar{B}| < \infty$. Otherwise $|A| < \infty$ and $|B| < \infty$, so $|A \cup B| < \infty$. So $A \cup B \in \mathcal{A}$

Thus $\mathcal{A}$ is a ring (algebra since $\mathbb{N} \in \mathcal{A}$)

(b) For each $n \in \mathbb{N}$, $\{2n\} \in \mathcal{A}$.

But $\bigcup_{n \in \mathbb{N}} \{2n\}$ is not since $|\bigcup_{n \in \mathbb{N}} \{2n\}| = \infty$,

and $\overline{\bigcup_{n \in \mathbb{N}} \{2n\}} = \bigcup_{n \in \mathbb{N}} \{2n+1\}$ and $|\bigcup_{n \in \mathbb{N}} \{2n+1\}| = \infty$

(c) Let $\{A_n\}_{n=1}^N \subseteq \mathcal{A}$ be p.d. with $A = \sum_{n=1}^N A_n \in \mathcal{A}$

If any $n \in \{1, 2, \ldots, N\}$ is such that $|A_n| = \infty$, then $|A| = \infty$ and we are done. Otherwise, $|A| < \infty$, and so $\mu(A) = 0$, $\sum_{n=1}^N \mu(A_n) = \sum_{n=1}^N 0 = 0$

(d) Let $A_n = \{n\}$. Then $\sum_{n=1}^\infty \mu(A_n) = \sum_{n=1}^\infty 0 = 0$

but $\sum_{n=1}^\infty A_n = \mathbb{N} \setminus \{0\}$ so $\mu\left(\sum_{n=1}^\infty A_n\right) = \infty > \sum_{n=1}^\infty \mu(A_n)$

Problem 8

① We know $\mathcal{A}$ is a σ-algebra

② $\mu(\emptyset) = \sum_{n=1}^{\infty} a_n \mu_n(\emptyset) = \sum_{n=1}^{\infty} 0 = 0$

③ $\mu\left(\sum_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} a_n \mu_n\left(\sum_{N=1}^{\infty} A_N\right)$

$= \sum_{n=1}^{\infty} a_n \left(\sum_{N=1}^{\infty} \mu_n(A_N)\right)$

$= \sum_{N=1}^{\infty} \sum_{n=1}^{\infty} a_n \mu_n(A_N)$

$= \sum_{n=1}^{\infty} \mu(A_N)$

So μ is a measure

Problem 9

(a) • $\mu^*(\emptyset) = 0$ by definition.

• Since $\emptyset \subseteq A \subseteq \Omega$ for any $A \subseteq \Omega$, and $\mu^*(\emptyset) \leq \mu^*(A) \leq \mu^*(\Omega)$ with $\mu^*(A) = 1$ for any $A \neq \emptyset$, $A \neq \Omega$, clearly $A \subseteq B \Rightarrow \mu^*(A) \leq \mu^*(B)$.
(If $A = \emptyset$, or $B = \Omega$, it is trivial, $A = B$ as well, ~~or~~ otherwise both $\mu^*(A) = \mu^*(B) = 1$)

• Let $A = \bigcup_{n=1}^{\infty} A_n$. If at least 2 A_i and A_j are non-empty then $\mu^*(A) \leq 2 \leq \sum_{n=1}^{\infty} \mu^*(A_n)$
Otherwise $A = A_i$ for some $i \in \mathbb{N} \setminus \{0\}$ and $\mu^*(A) = \mu^*(A_i) = \sum_{n=1}^{\infty} \mu^*(A_n)$ (everything else is $\emptyset$)

So μ^* is an outer measure

(b) $\mu^*(\Omega) = 2$, but $\mu^*(\{1\}) + \mu^*(\{2\}) + \mu^*(\{3\}) = 3$

(c) We seek all A such that
$\mu^*(A \cap E) + \mu^*(\bar{A} \cap E) = \mu^*(E)$ for all $E \subseteq \Omega$
But $\mu^*(\emptyset) = 0$, so we need $\mu^*(A \cap E) = \mu^*(\bar{A} \cap E)$
$= 0$. For $E = \Omega$, we need $A = \bar{A} = \emptyset$ which is
a contradiction.
So $\mathcal{M}(\mu^*) = \emptyset$